

Simulating Markov chains with transient and recurrent states



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Abstract

Numerical simulations of a Markov chain with a mixture of transient and recurrent states have been performed. The values obtained by sampling are compared with the values for these quantities that can be obtained using derivations based on the partition theorem and conditional expectation theorems. The values obtained using these formulas are shown to be within the error bars on the estimates of these quantities that are obtained by sampling. The two methods are thus estimating the same quantities.

1 Introduction

I chose to simulate the Markov chain with the transition graph shown in figure 1 for my project. The transition matrix for this graph is as follows:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1/3 & 1/3 & 1/3 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 1.0 \end{pmatrix}$$

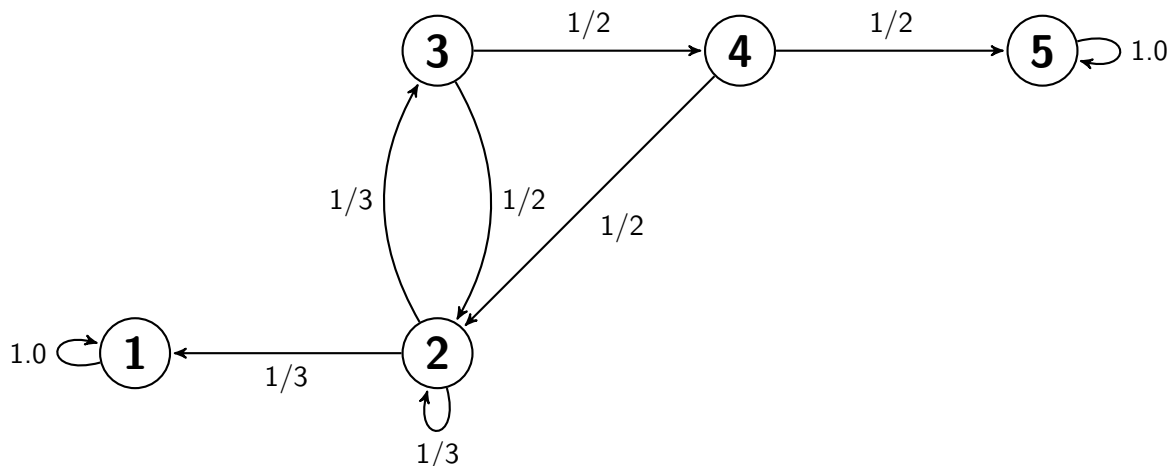


Fig. 1. The transition graph that was simulated for this report.

States 2, 3 and 4 in this chain are transient and states 1 and 5 are recurrent and absorbing. In section 2 I will explain how I estimated the hitting probabilities for this chain by sampling and how these quantities were calculated using matrix algebra. In section 3 I will perform similar calculations to determine the hitting times.

2 Hitting probabilities

A function was written to sample the Markov chain illustrated in figure 1. The initial state was one of the expected arguments for this function. This initial state was expected to be one of the transient states, i.e., state 2, 3 or 4. The function then simulated the Markov chain and returned the final absorbing state (state 1 or 5) into which the chain was absorbed.

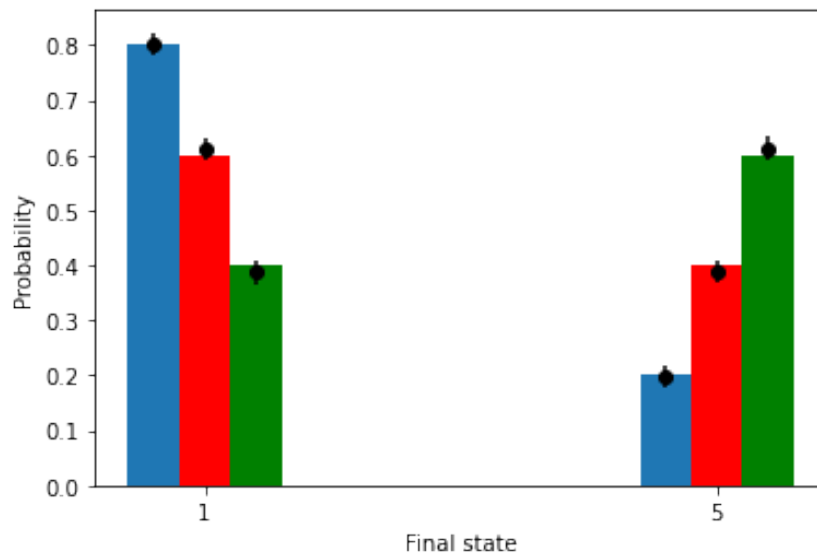


Fig. 2. Hitting probabilities for the Markov chain shown in figure 1. The blue, red and green bars show the conditional probabilities of finishing in the two recurrent states, given that you start in states 2, 3 and 4, respectively. The heights of these bars show the values obtained for these quantities from the analytic formula. The black error bars show the values of these quantities obtained by sampling together with an error bar that illustrates the 90% confidence limit on this estimate.

The function described in the previous paragraph was run starting from each of the transient states 1000 times, and the number of times that the run ended in state one was accumulated. The resulting estimates of the hitting probabilities, together with an error bar that represents a 95% confidence limit, are shown in figure 2.

We can also calculate the hitting probabilities exactly by using the following expression:

$$\mathbf{H} = (\mathbf{I} - \mathbf{Q})^{-1}\mathbf{R}$$

In this expression, $\mathbf{I}$ is the identity matrix, and $\mathbf{Q}$ and $\mathbf{R}$ are the following matrices:

$$\mathbf{Q} = \begin{pmatrix} 1/3 & 1/3 & 0 \\ 1/2 & 0 & 1/2 \\ 1/2 & 0 & 0 \end{pmatrix} \quad \text{and} \quad \mathbf{R} = \begin{pmatrix} 1/3 & 0 \\ 0 & 0 \\ 0 & 1/2 \end{pmatrix}$$

The elements of the hitting matrix that are obtained in this way are illustrated using bars in figure 2. The blue bars are the conditional probabilities of finishing in states 1 and 5, given you start in state 2. The red bars are the conditional probabilities of finishing in states 1 and 5, given you start in state 3. The green bars are the conditional probabilities of finishing in states 1 and 5, given you start in state 4. You can see that the values that you obtain using these expressions are within the error bars of the conditional probabilities that you obtain by sampling. You can thus be confident that the two methods described above are estimating the same quantities.

3 Hitting times

The python function described in section 2 was also used to estimate the hitting times. The hitting time measures the average number of steps before the chain arrives in one of the two absorbing states. We can thus calculate hitting times starting from each of the three transient states shown in figure 1.

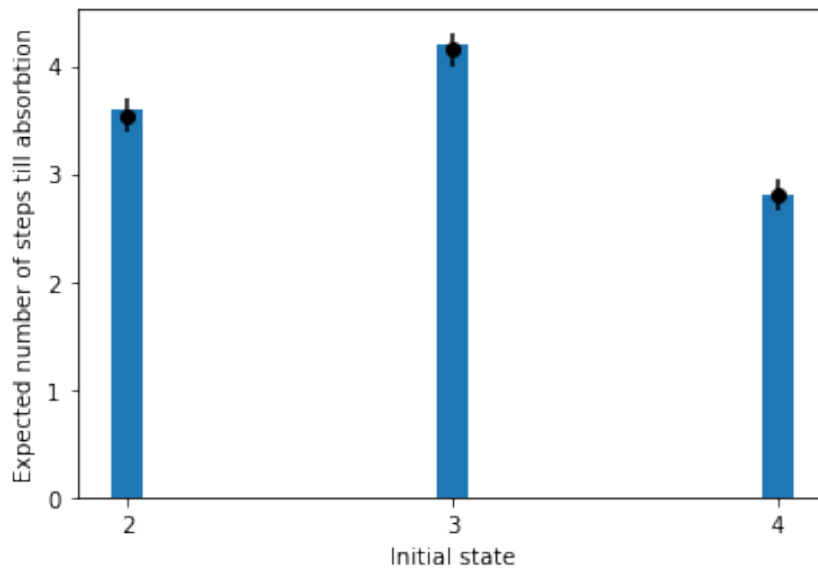


Fig. 3. Hitting times for the Markov chain shown in figure 1. The heights of the bars illustrate the expectations that are calculated using the analytic formula for this quantity. The black points show the values for these quantities that were obtained by sampling. The error bars illustrate a 90% confidence limit on the estimates of these quantities.

To estimate the hitting times by sampling the function described in 2 was run starting from each of the transient states 1000 times. The mean and the variance for the number of steps in each of these 1000 runs were then used to estimate the hitting times that are shown as black

dots in figure 3. A 95% confidence limit on this estimate was also computed and is illustrated using an error bar.

The hitting times can also be calculated using the following expression:

$$\mathbf{h} = (\mathbf{I} - \mathbf{Q})^{-1}\mathbf{1}$$

In this expression, $\mathbf{I}$ and $\mathbf{Q}$ are defined as they were in section 2. $\mathbf{1}$, meanwhile, is a column vector with three elements that are all equal to one. The hitting times that are obtained using this expression are illustrated in figure 3 by the bars. You can see that the heights of the bars fall within the error bars on the numerical estimates of the hitting times. We can thus safely conclude that these two methods are estimating the same quantity.